Combined Science: Chemical Reactions

IGCSE Worked Solutions

Each answer below is written using only the concepts taught in this chapter. If a question needs material beyond the chapter, it is flagged for teacher review.

Question 1. *[Combined Science 2025 11]*

21 Which description explains why a reaction involves reduction?

A mass decreases

B oxygen is lost

C pH decreases

D temperature decreases

22 Which number determines the order of elements in the Periodic Table?

Final answer: B. oxygen is lost

1. The question asks for the description that explains why a reaction involves REDUCTION. In the oxygen view taught in this chapter, reduction is defined as the LOSS of oxygen by a substance during a reaction. [from: Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.]

2. Check each option against the chapter definitions. A 'mass decreases' is not a definition of reduction (e.g. a thermal decomposition loses mass but is not always a reduction). C 'pH decreases' describes an acid being added, not redox. D 'temperature decreases' is about an endothermic change, not redox. [from: C6.03 Redox reactions]

3. Option B 'oxygen is lost' matches the chapter definition of reduction exactly (OIL RIG; oxygen-view: reduction = loss of oxygen). Therefore the correct answer is B. [from: OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.]

From your chapter: C6.03 Redox reactions; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.

Question 2. *[Combined Science 2025 11]*

23 Which statements about transition elements are correct?

1 They are in Group I of the Periodic Table.

2 They form coloured compounds.

3 They often act as catalysts.

4 They have low melting points.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

Final answer: C — 2 and 3 (transition elements form coloured compounds and often act as catalysts).

1. Check statement 2: 'They form coloured compounds.' This is TRUE. You have met coloured transition-metal compounds in this chapter — blue copper(II) sulfate solution ($CuSO_4$) in the zinc + copper sulfate displacement, and iron compounds are used as examples with Roman numerals (iron(II), iron(III)). So statement 2 is correct. [from: C6.03 Redox reactions (transition metal, Fe^{2+}, Fe^{3+}, Cu^{2+}, copper(II) sulfate)]

2. Check statement 3: 'They often act as catalysts.' This is TRUE. A catalyst speeds up a reaction by lowering the activation energy and is unchanged at the end. A clear chapter

example is manganese(IV) oxide (MnO_2), a transition-metal oxide, which catalyses the decomposition of hydrogen peroxide: $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$. So statement 3 is correct. [from: C6.02 Rate of reaction (catalyst, MnO_2, hydrogen peroxide decomposition)]

3. Check statement 1: 'They are in Group I of the Periodic Table.' This is FALSE. Group I contains the alkali metals (Li, Na, K, ...). Transition elements sit in the middle block of the Periodic Table, between Group II and Group III, not in Group I. [from: C6.03 Redox reactions (transition metal terminology)]

4. Check statement 4: 'They have low melting points.' This is FALSE. Transition elements are typical metals with HIGH melting points (for example iron melts at about 1538 degrees C, copper at about 1085 degrees C). The alkali metals in Group I are the ones with unusually low melting points for metals. [from: C6.03 Redox reactions (transition metal terminology)]

5. Only statements 2 and 3 are correct, so the answer is option C. [from: C6 elimination]

From your chapter: C6.02 Rate of reaction; C6.03 Redox reactions; catalyst; manganese(IV) oxide; MnO_2; hydrogen peroxide; transition metal; Cu^{2+}; Fe^{2+}; Fe^{3+}; Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.; Decomposition of hydrogen peroxide (catalysed by MnO_2): $2H_2O_2(l) \rightarrow 2H_2O(l) + O_2(g)$.

Question 3. *[Combined Science 2025 13]*

A Heat liquid water to form steam.

B Mix powdered carbon and iron filings.

C Melt iron filings.

D Heat sulfur in air to make sulfur dioxide.

Final answer: D (heating sulfur in air to make sulfur dioxide is a chemical change because a new substanc

1. The question asks which process results in a chemical change. The key test from the chapter is: a chemical change forms one or more NEW substances, while a physical change only changes state, form or appearance with no new substance. [from: C6.01 Physical and chemical changes]

2. Option A - Heating liquid water to form steam. Liquid water (H_2O) turns into gaseous water (H_2O). Only the state has changed; it is still water. No new substance is formed, so this is a physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

3. Option B - Mixing powdered carbon and iron filings. Each grain of carbon is still carbon and each piece of iron is still iron; you could separate them again with a magnet. No new substance is formed, so this is only a physical mixing, not a chemical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

4. Option C - Melting iron filings. Solid iron becomes liquid iron, but it is still iron. Only the state has changed, so this is a physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

5. Option D - Heating sulfur in air to make sulfur dioxide. Sulfur (S) reacts with oxygen (O_2) to form a brand new substance, sulfur dioxide (SO_2). This is combustion, which is a chemical

reaction and is exothermic. A new substance is formed, so this IS a chemical change. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

6. Therefore the correct answer is D. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.

Question 4. *[Combined Science 2025 22]*

and water.

What is the balanced symbol equation and correct state symbols for this reaction?

A $CaCO_3(s) + HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

B $CaCO_3(s) + HCl(l) \rightarrow CaCl_2(s) + CO_2(g) + H_2O(aq)$

C $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

D $CaCO_3(s) + 2HCl(l) \rightarrow CaCl_2(s) + CO_2(g) + H_2O(aq)$

Final answer: C — CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + CO2(g) + H2O(l)

1. Write the word equation from the stem: calcium carbonate + hydrochloric acid -> calcium chloride + carbon dioxide + water. [from: Calcium carbonate + hydrochloric acid: CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + H2O(l) + CO2(g).]

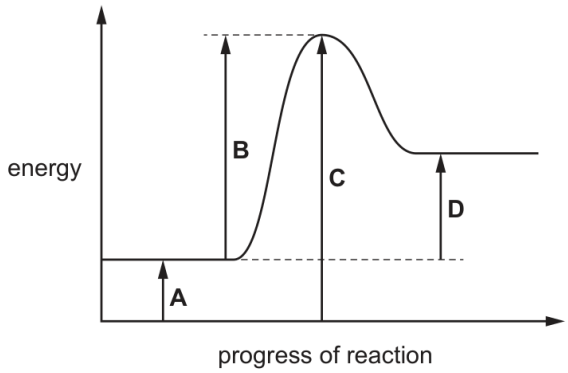
2. Balance the atoms. CaCl2 has 2 Cl atoms, so we need 2 HCl on the left. That also gives 2 H atoms on the left, which balances the 2 H atoms in H2O on the right. Ca: 1 = 1. C: 1 = 1 (in CO2). O: 3 on the left (in CaCO3) = 2 (CO2) + 1 (H2O) = 3 on the right. Balanced equation: CaCO3 + 2HCl -> CaCl2 + CO2 + H2O. [from: Calcium carbonate + hydrochloric acid: CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + H2O(l) + CO2(g).]

3. Add correct state symbols. CaCO3 is a solid (s) — it is the marble chip. Dilute hydrochloric acid is an aqueous solution, so HCl is (aq). Calcium chloride dissolves in the acid, so CaCl2 is (aq). Carbon dioxide is a gas, so CO2 is (g). Water formed is a liquid, so H2O is (l). [from: C6.02 Rate of reaction (marble chips + dilute HCl)]

4. Compare with the options. Option A is not balanced (only 1 HCl). Option B has HCl(l) and CaCl2(s) and H2O(aq), all wrong states. Option D also has HCl(l) and CaCl2(s), wrong states. Option C matches exactly: CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + CO2(g) + H2O(l). Answer is C. [from: Calcium carbonate + hydrochloric acid: CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + H2O(l) + CO2(g).]

From your chapter: C6.02 Rate of reaction; calcium carbonate; hydrochloric acid; marble chips; Calcium carbonate + hydrochloric acid: CaCO3(s) + 2HCl(aq) -> CaCl2(aq) + H2O(l) + CO2(g).

Question 5. *[Combined Science 2025 22]*



Final answer: B (the arrow from the reactants line up to the top of the hump shows the activation energy Ea

1. Identify the reaction profile shown. The left flat line is labelled 'reactants'. The curve rises over a hump (the peak) and then drops to a higher flat line on the right labelled 'products'. Because products are HIGHER than reactants, this is an endothermic reaction profile (energy is taken in from the surroundings). [from: C6.01 Physical and chemical changes (endothermic: energy taken in, products higher than reactants)]

2. Recall the definition of activation energy (Ea): it is the minimum energy that colliding particles must have for a reaction to occur. On a reaction pathway diagram, Ea is the vertical height of the 'hump' measured from the REACTANTS line UP to the top of the peak (the highest point of the curve). [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

3. Check each labelled arrow on the diagram: A points from the x-axis up to the reactants line (the energy of the reactants measured from zero, NOT Ea). B points from the reactants line UP to the top of the peak, so B IS the activation energy. C points from the x-axis up to the top of the peak (the total energy of the peak from zero, NOT Ea). D points from the reactants dashed line up to the products line (the overall energy change, delta H, for this endothermic reaction, NOT Ea). [from: C6.02 Rate of reaction (reaction pathway diagram with activation energy Ea as the hump)]

4. Only arrow B matches the definition 'height of the hump measured from the reactants line up to the peak', so the activation energy for this endothermic reaction is labelled B. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; reaction pathway diagram; activation energy; Ea; endothermic; exothermic

Question 6. *[Combined Science 2025 12]*

20 Which statement indicates that a chemical change occurs?

A Energy is absorbed.

B There is a change in colour.

C There is a change of state.

D There is no change in mass.

Final answer: B (a change in colour indicates a chemical change because it suggests a new substance has

1. Recall the rule: a chemical change (chemical reaction) is one in which one or more new substances are formed. Signs that new substances have formed include a colour change, a precipitate forming, a gas being produced, or a change in energy that is not simply a change of state. [from: C6.01 Physical and chemical changes]

2. Check option A (energy is absorbed). Energy changes happen in both physical changes (e.g. ice melting absorbs heat) and chemical changes (endothermic reactions), so 'energy absorbed' alone does NOT prove a chemical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

3. Check option B (there is a change in colour). A colour change is one of the classic clues listed for a chemical reaction, because a different-coloured substance means a new substance has probably been made. This matches the definition of a chemical change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

4. Check option C (there is a change of state). A change of state such as melting, freezing, boiling or condensing is the textbook example of a PHYSICAL change, not a chemical one. [from: C6.01 Physical and chemical changes]

5. Check option D (there is no change in mass). Mass is conserved in every reaction in a closed system, whether physical OR chemical, so 'no change in mass' does not tell you which type of change occurred. [from: C6.01 Physical and chemical changes]

6. Only option B is a reliable indicator of a new substance forming, so the answer is B. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.

Question 7. *[Combined Science 2025 32]*

6 Hydrogen peroxide, H_2O_2, decomposes into water and oxygen when a catalyst is added.

(a) Balance the symbol equation for the reaction.

$$\ldots\ldots\ldots\ldots H_2O_2 \rightarrow \ldots\ldots\ldots\ldots H_2O + O_2$$

[

(b) The oxygen is collected and tested.

Describe the test for oxygen gas.

State the observation for the positive result.

test .....

observation .....

[

(c) Anhydrous cobalt(II) chloride is used to test for water.

State the colour observed for a positive result.

..... [

(d) Describe what is meant by a catalyst.

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..... [

(e) The temperature of the hydrogen peroxide is increased. All other conditions stay the same.

Predict the effect of this change on the reaction.

..... [

[Total:

Final answer: (a) 2H2O2 -> 2H2O + O2; (b) a glowing splint placed at the mouth of the tube relights; (c) pi

1. (a) The unbalanced equation is H2O2 -> H2O + O2. Count atoms: left side has 2H and 2O; right side has 2H and 3O, so oxygen is not balanced. Put a 2 in front of H2O2 and a 2 in front of H2O: 2H2O2 -> 2H2O + O2. Now both sides have 4 H atoms and 4 O atoms, so the equation is balanced. [from: Decomposition of hydrogen peroxide (catalysed by MnO2): 2H2O2(l) -> 2H2O(l) + O2(g).]

2. (b) The gas given off in this reaction is oxygen (O2). The standard school test for oxygen gas is to place a GLOWING (not burning) wooden splint at the mouth of the test tube. The positive observation is that the glowing splint RELIGHTS, showing that the gas is oxygen. [from: Decomposition of hydrogen peroxide (catalysed by MnO2): 2H2O2(l) -> 2H2O(l) + O2(g).]

3. (c) Anhydrous cobalt(II) chloride starts blue when dry. When water touches it, it turns PINK (it becomes the hydrated form). So the colour observed for a positive result (water present) is pink. [from: Decomposition of hydrogen peroxide (catalysed by MnO2): 2H2O2(l) -> 2H2O(l) + O2(g).]

4. (d) A catalyst is a substance that INCREASES the rate of a chemical reaction (point 1 of 2). It does this by providing an alternative reaction pathway with a LOWER activation energy (Ea), so a larger fraction of collisions are successful. Also, the catalyst is CHEMICALLY UNCHANGED at the end of the reaction - you can recover the same mass of it afterwards (point 2 of 2). [from: Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.]

5. (e) Increasing the temperature gives the H2O2 particles more kinetic energy, so they move faster and collide more often, AND a greater proportion of the collisions now have energy >= Ea (this second effect is the bigger one). So the rate of decomposition increases - the reaction is FASTER and finishes sooner. [from: Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).]

From your chapter: Decomposition of hydrogen peroxide (catalysed by MnO2): 2H2O2(l) -> 2H2O(l) + O2(g).; Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.; Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).; C6.02 Rate of reaction

Question 8. *[Combined Science 2025 21]*

22 Which processes involve a chemical change?

1 cracking of an alkane

2 polymerisation of ethene

3 corrosion of iron

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

Final answer: A (1, 2 and 3 - cracking, polymerisation and corrosion all form new substances, so all three a

1. The chapter rule: a process is a chemical change if one or more NEW substances are formed. If only the state or form changes and the same substance remains, it is a physical change. Check each of the three processes for a new substance. [from: C6.01 Physical and chemical changes]

2. Process 1 - Cracking of an alkane. Cracking breaks a long alkane molecule into smaller molecules (a smaller alkane plus an alkene). The starting molecule is gone and two different molecules are made - so new substances are formed. This IS a chemical change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

3. Process 2 - Polymerisation of ethene. Many ethene molecules (C2H4) join together to make a long polymer chain (poly(ethene)). The product is a completely different substance from the reactant, so a new substance has been formed. This IS a chemical change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

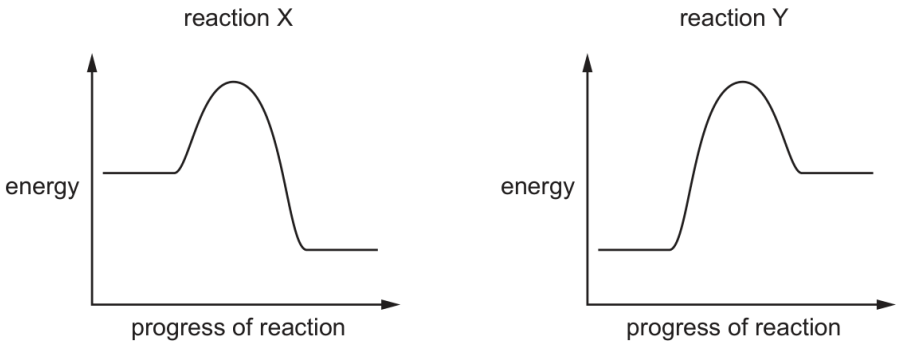
4. Process 3 - Corrosion of iron. When iron corrodes it reacts with oxygen (and water) to form iron oxide (rust). Iron has gained oxygen, so this is oxidation - a new substance (iron oxide) is formed. Corrosion IS a chemical change (and in particular a redox reaction). [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

5. All three processes form new substances, so all three are chemical changes. The answer is A (1, 2 and 3). [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; C6.03 Redox reactions

Question 9. *[Combined Science 2025 21]*

20 The reaction pathway diagrams for reaction X and for reaction Y are shown.



Which statement about the reactions is correct?

A Reaction X has a greater activation energy than reaction Y.

B Reaction X is endothermic and reaction Y is exothermic.

C The overall energy change in reaction X is much greater than in reaction Y.

D The temperature increases during reaction X and decreases during reaction Y.

Final answer: D (the temperature increases during reaction X and decreases during reaction Y)

1. Read reaction X's diagram. The reactants line is HIGH on the energy axis, the curve rises to a peak, then falls DOWN to a lower products line. Because products are LOWER in energy than reactants, reaction X gives energy out overall. That makes X an exothermic reaction, so the surroundings get hotter and the temperature increases during reaction X. [from: C6.01 Physical and chemical changes]

2. Read reaction Y's diagram. The reactants line is LOW, the curve rises to a peak, then falls only slightly to a products line that is still HIGHER than the reactants. Because products are HIGHER in energy than reactants, reaction Y takes energy in overall. That makes Y an endothermic reaction, so energy is taken from the surroundings and the temperature decreases during reaction Y. [from: C6.01 Physical and chemical changes]

3. Option A says reaction X has a GREATER activation energy than reaction Y. The activation energy is the height of the hump measured from the reactants line up to the top of

the peak. In these two diagrams the humps look about the same size, so we cannot say X is greater. A is not correct. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

4. Option B says X is endothermic and Y is exothermic. This is the wrong way round. X is exothermic (products lower than reactants) and Y is endothermic (products higher). B is not correct. [from: C6.01 Physical and chemical changes]

5. Option C says the overall energy change in X is much greater than in Y. The overall energy change is the vertical distance between the reactants line and the products line. In the two diagrams the drops/rises look similar in size, so we cannot claim X's energy change is much greater. C is not correct. [from: C6.01 Physical and chemical changes]

6. Option D says the temperature increases during reaction X and decreases during reaction Y. This matches steps 1 and 2: X is exothermic (temperature of surroundings rises) and Y is endothermic (temperature of surroundings falls). D is correct. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.

Question 10. *[Combined Science 2025 21]*

A addition

B combustion

C displacement

D polymerisation

Final answer: A — addition

1. First, check this really is a chemical reaction. Alkenes have a C=C double bond. When they react with bromine (Br_2), a brand-new substance called 1,2-dibromoalkane is made, and the orange bromine solution turns colourless. Because a new substance is formed, this counts as a chemical change (a chemical reaction) — the question is just asking which TYPE of chemical reaction it is. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed]

2. Rule out option B, combustion. From our chapter, combustion means a substance reacts with oxygen and gives out energy (like methane + O_2). In this reaction there is no oxygen anywhere — only alkene + bromine — so it cannot be combustion. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

3. Rule out option C, displacement. Our chapter says displacement is when a more reactive element pushes a less reactive element out of its compound or solution (like $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$). Here, nothing is being 'kicked out' — both bromine atoms end up inside the new product molecule, and no element is being displaced from a compound. So it is not displacement. [from: Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.]

4. Rule out option D, polymerisation. Polymerisation would join many alkene molecules together into one very long chain. Here only one alkene molecule reacts with one Br_2 molecule to make one small product — no long chain is formed — so it is not polymerisation.

[from: Chemical change (chemical reaction): a change in which one or more new substances are formed]

5. The only remaining option is A, addition. The two bromine atoms 'add on' across the C=C double bond: the double bond becomes a single bond, and one Br attaches to each carbon, forming 1,2-dibromoalkane. No atoms are lost — everything that went in is in the single new product. This fits the chapter rule that a chemical reaction makes a new substance. (This colour change, orange → colourless, is the classic school test for an alkene.) [from: Chemical change (chemical reaction): a change in which one or more new substances are formed]

From your chapter: C6.01 Physical and chemical changes; Chemical change (chemical reaction): a change in which one or more new substances are formed; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.; Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.

Question 11. *[Combined Science 2025 11]*

A condensation

B evaporation

C melting

D oxidation

Final answer: D (oxidation is a chemical change because a new substance is formed when oxygen is gaine

1. The question asks which process involves a chemical change. The rule from the chapter: a chemical change forms one or more NEW substances; a physical change only changes the state or appearance of the same substance. [from: C6.01 Physical and chemical changes]

2. Option A - Condensation. A gas turns into a liquid (e.g. steam turning back to liquid water). The substance itself is the same, only its state changes, so this is a physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

3. Option B - Evaporation. A liquid turns into a gas (e.g. water becoming water vapour). No new substance is made, so this is also a physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

4. Option C - Melting. A solid turns into a liquid (e.g. ice melting to water). Same substance, different state, so this is another physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

5. Option D - Oxidation. In oxidation a substance gains oxygen (or loses electrons) and a NEW substance is formed. For example, magnesium + oxygen -> magnesium oxide, or iron rusting to form iron oxide. A new substance appears, so oxidation is a chemical change. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

6. Therefore the correct answer is D. [from: C6.03 Redox reactions]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; C6.03 Redox reactions

Question 12. *[Combined Science 2025 11]*

The word equation for the complete combustion of pentane is shown.

pentane + oxygen → carbon dioxide + substance W

What is substance W?

A carbon monoxide

B ethane

C hydrogen

D water

Final answer: D water

1. Pentane is a hydrocarbon, which means it only contains carbon and hydrogen atoms. When any hydrocarbon burns completely in plenty of oxygen (complete combustion), the two products are always carbon dioxide (CO_2) and water (H_2O). [from: Combustion of methane: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (exothermic, redox).]

2. The equation given is: pentane + oxygen -> carbon dioxide + substance W. Carbon dioxide is already listed, so W must be the other product of complete combustion, which is water. [from: C6.03 Redox reactions]

3. Check the other options. A 'carbon monoxide' would form only in incomplete combustion (not enough oxygen), but the question says complete combustion. B 'ethane' is another hydrocarbon fuel, not a product. C 'hydrogen' is not a product of burning - the hydrogen atoms in pentane bond with oxygen to make water. So the correct answer is D, water. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

From your chapter: C6.03 Redox reactions; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.; Combustion of methane: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (exothermic, redox).

Question 13. *[Combined Science 2025 42]*

5 **(a)** Calcium oxide, CaO, is used to make cement.

Calcium oxide is formed by the decomposition of calcium carbonate, $CaCO_3$, at a very high temperature.

The equation for the reaction is shown.

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

(i) Suggest why making cement contributes to global warming.

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..... [2]

(ii) The reaction is endothermic.

Explain what is meant by an endothermic reaction.

Use ideas about making bonds and breaking bonds in your answer.

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..... [3]

Final answer: (a)(i) Making cement releases CO2 in two ways: (1) the decomposition itself CaCO3(s) -> Ca

1. (a)(i) Step 1: Look at the chemical equation. CaCO3(s) -> CaO(s) + CO2(g). The decomposition of calcium carbonate itself produces carbon dioxide (CO2) as one of the products. [from: decomposition (C6.01) / CaCO3 reactions in formulas]

2. (a)(i) Step 2: The stem says 'at a very high temperature'. To reach that temperature, fuel has to be burnt (combustion). Combustion of fuels (e.g. CH4(g) + 2O2(g) -> CO2(g) + 2H2O(g), an exothermic combustion in the profile) also releases CO2. [from: Combustion of methane: CH4(g) + 2O2(g) -> CO2(g) + 2H2O(g) (exothermic, redox).]

3. (a)(i) Step 3: Carbon dioxide is a greenhouse gas. Releasing extra CO2 into the atmosphere increases the greenhouse effect and contributes to global warming. Two mark-bearing ideas: (1) CO2 is produced (by the decomposition and/or by burning fuel to heat the kiln) and (2) CO2 is a greenhouse gas / traps heat / causes the greenhouse effect. [from: C6.01 Physical and chemical changes (new substance produced: CO2)]

4. (a)(ii) Step 1: State the definition. An endothermic reaction is a chemical reaction in which energy (heat) is taken in (absorbed) FROM the surroundings. The surroundings cool down. [from: C6.01 Physical and chemical changes (energy can be given out (exothermic) or taken in (endothermic))]

5. (a)(ii) Step 2: Bond breaking is an energy-IN process - to break the strong bonds in the reactants (the bonds in $CaCO_3$), energy must be supplied. Bond making is an energy-OUT process - when new bonds form in the products (in CaO and CO_2), energy is released. [from: C6.01 Physical and chemical changes (chemical reaction, energy change)]

6. (a)(ii) Step 3: Compare the two energies. In an endothermic reaction the energy required to break the reactant bonds is GREATER than the energy released when the new product bonds form. The difference is taken in from the surroundings, which is why the overall reaction is endothermic. This is exactly why $CaCO_3$ has to be heated strongly in a kiln to decompose. [from: C6.01 exothermic/endothermic + chemical change definition]

From your chapter: C6.01 Physical and chemical changes; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.; Combustion of methane: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (exothermic, redox).; decomposition; calcium carbonate

Question 14. *[Combined Science 2025 22]*

18 Which statement explains why the rate of a reaction is greater at a higher temperature?

A The activation energy is higher.

B More colliding particles have the minimum energy to react.

C The frequency of collisions between reacting particles decreases.

D There are more reacting particles per unit volume.

Final answer: B (at higher temperature more colliding particles have at least the minimum energy - the acti

1. The question asks why RATE is greater at a HIGHER temperature. Use collision theory: a reaction only happens when particles collide with energy >= Ea (activation energy). [from: Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.]

2. Check option A: 'The activation energy is higher.' Ea is a property of the reaction itself and does NOT change when you warm the mixture. Even if it did go up, that would make the reaction slower, not faster. Wrong. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

3. Check option B: 'More colliding particles have the minimum energy to react.' At higher T particles have more kinetic energy, so a GREATER PROPORTION of collisions have energy >= Ea. More of the collisions are SUCCESSFUL, so rate goes up. This matches the chapter exactly. Correct. [from: Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).]

4. Check option C: 'The frequency of collisions between reacting particles decreases.' At higher T particles move faster and collide MORE often, not less. Also, a decrease in collision frequency would SLOW the reaction, not speed it up. Wrong. [from: C6.02 Rate of reaction]

5. Check option D: 'There are more reacting particles per unit volume.' Heating a liquid or gas does not add more particles; the number of particles stays the same (it can even slightly decrease particles per unit volume if the mixture expands). That describes a concentration / pressure change, not a temperature change. Wrong. [from: C6.02 Rate of reaction]

6. Only option B gives the correct collision-theory reason for a faster reaction at higher temperature. Answer = B. [from: Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.]

From your chapter: C6.02 Rate of reaction; Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).

Question 15. *[Combined Science 2025 22]*

24 Which process is a physical change?

A combustion of solid carbon

B cracking of an alkane to give an alkene

C fractional distillation of petroleum

D rusting of iron

Final answer: C (fractional distillation of petroleum — no new substances are formed, the hydrocarbons are

1. Recall the key idea. A physical change does NOT make any new substance — it only changes the state, form or situation (e.g. melting, boiling, evaporating, dissolving, separating a mixture). A chemical change makes one or more new substances. [from: C6.01 Physical and chemical changes]

2. Test option A: combustion of solid carbon. Carbon burns in oxygen to form a new substance, carbon dioxide ($C + O_2 \rightarrow CO_2$). A new substance is formed, so combustion is a CHEMICAL change. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

3. Test option B: cracking an alkane to give an alkene. Cracking breaks a big hydrocarbon molecule into smaller, different molecules (for example long-chain alkane -> shorter alkane + alkene). The products are different substances from the reactant, so this is a CHEMICAL change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

4. Test option C: fractional distillation of petroleum. Petroleum is a mixture of hydrocarbons with different boiling points. Fractional distillation just heats the mixture so each hydrocarbon boils off and is condensed in turn. The same hydrocarbon molecules come out the other end — no new substance is formed. So this is a PHYSICAL change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

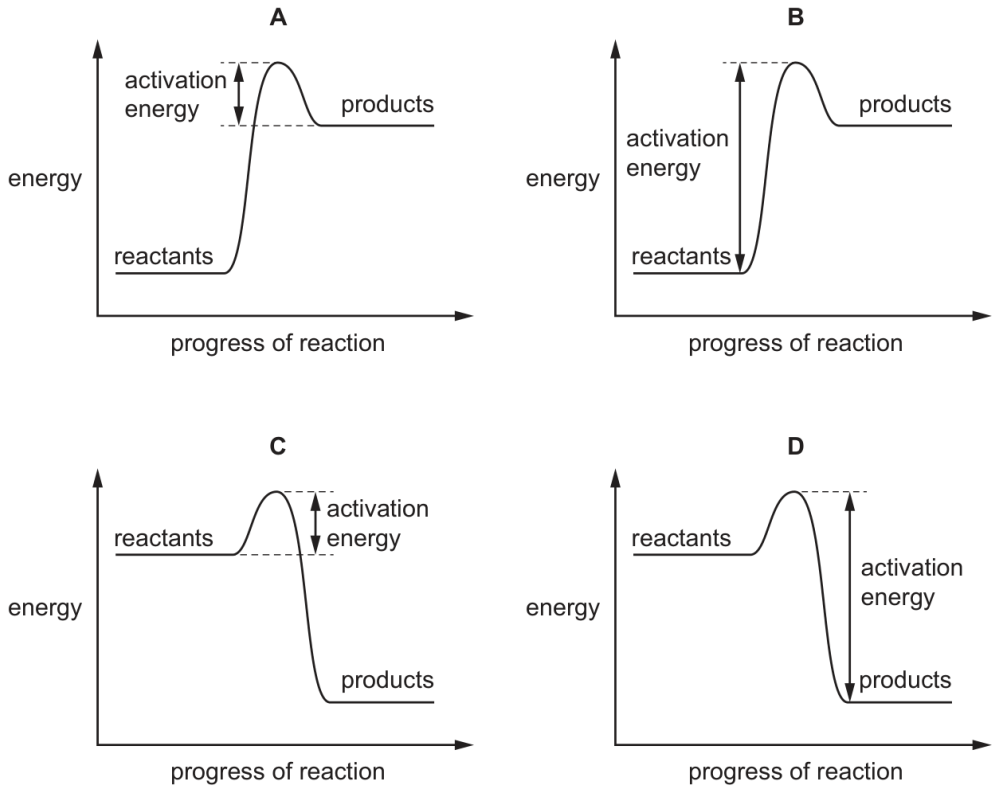
5. Test option D: rusting of iron. Iron reacts with oxygen (and water) to form a new substance, iron oxide (rust). A new substance is formed, so rusting is a CHEMICAL change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

6. Only option C does not produce a new substance, so the answer is C. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.

Question 16. *[Combined Science 2025 22]*

18 Which reaction pathway diagram shows the activation energy for an exothermic reaction?



Final answer: C (the exothermic reaction profile where reactants are higher than products and the activatio

1. Recall what an exothermic reaction profile looks like: the products line must be LOWER than the reactants line, because the reaction gives out energy to the surroundings. That rules out any diagram where products are shown above reactants. [from: C6.01 Physical and chemical changes (exothermic: energy given out, products lower than reactants)]

2. Recall the definition of activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur. On a reaction pathway diagram Ea is the height of the 'hump' measured from the REACTANTS line UP to the top of the peak, not from the products line. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

3. Check each of the four diagrams: Diagram A shows reactants LOWER than products (endothermic, wrong profile for this question). Diagram B also shows reactants LOWER than products (endothermic, wrong profile). Diagram C shows reactants HIGHER than products (exothermic - correct profile), and the arrow labelled 'activation energy' runs from the reactants dashed line UP to the top of the peak (correct definition of Ea). Diagram D shows an exothermic profile, but the arrow labelled 'activation energy' runs from the PRODUCTS line up to the peak, which is too large - that would be the reverse Ea, not the forward Ea. [from: C6.02 Rate of reaction (reaction pathway diagram; Ea as height of hump above reactants)]

4. Only diagram C shows BOTH an exothermic reaction (products lower than reactants) AND the activation energy correctly drawn from the reactants line up to the peak. So the answer is C. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; reaction pathway diagram; activation energy; Ea; exothermic; endothermic

Question 17. *[Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

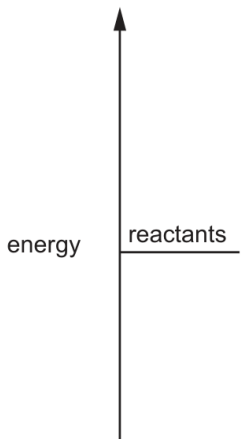
A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.



5 The reaction between aluminium and iron(III) oxide is used to join iron railway lines together.

$$2Al(s) + Fe_2O_3(s) \rightarrow 2Fe(l) + Al_2O_3(s)$$

Molten iron is produced in the reaction. Molten iron becomes solid when it cools.

(a) **(i)** State how the equation shows that **molten** iron is produced.

..... [1]

(ii) Use the equation to explain why this reaction is a redox reaction.

.....

.....

..... [2]

(iii) Complete the ionic equation for this reaction.

$Al + Fe^{3+} \rightarrow$ ..... + ... [2]

(b) Explain the meaning of the symbol (III) in iron(III) oxide.

.....

..... [1]

Final answer: (a)(i) The state symbol (l) after Fe means iron is a liquid (molten). (a)(ii) Aluminium gains oxy

1. (a)(i) Look at the state symbols in the equation 2Al(s) + Fe2O3(s) -> 2Fe(l) + Al2O3(s). The (l) written after Fe means LIQUID. So iron is produced as a liquid (molten) in this reaction. That is the clue the equation gives. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

2. (a)(ii) Use the oxygen definition of redox. Aluminium (Al) starts with no oxygen and ends up in Al2O3, so aluminium has GAINED oxygen - aluminium is OXIDISED. Iron(III) oxide (Fe2O3) starts with oxygen and ends up as plain Fe, so Fe2O3 has LOST oxygen - Fe2O3 is REDUCED. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

3. (a)(ii) Because oxidation and reduction both happen in the same reaction (Al is oxidised AT THE SAME TIME as Fe2O3 is reduced), this is a REDOX reaction. That is the definition of redox. [from: Redox reaction: a reaction involving simultaneous reduction and oxidation.]

4. (a)(iii) Complete the ionic equation Al + Fe3+ -> ? + ?. Aluminium is a metal that forms Al3+ ions (lost 3 electrons - oxidation). Fe3+ gains 3 electrons to become neutral Fe atoms -

reduction. So the products are Al3+ and Fe. [from: OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.]

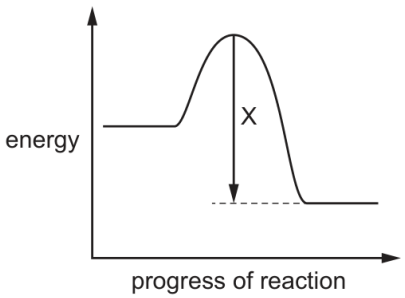
5. (a)(iii) Check electrons balance: Al -> Al3+ + 3e- (loses 3 electrons) and Fe3+ + 3e- -> Fe (gains 3 electrons). Three electrons leave Al and three arrive at Fe3+, so electrons balance 1:1. The completed ionic equation is Al + Fe3+ -> Al3+ + Fe. [from: Displacement: Zn(s) + CuSO4(aq) -> ZnSO4(aq) + Cu(s); ionic Zn + Cu2+ -> Zn2+ + Cu (Zn oxidised, Cu2+ reduced).]

6. (b) The (III) in iron(III) oxide is a ROMAN NUMERAL. In chemistry it is used to show the oxidation number (charge on the ion) of the transition metal. So iron(III) means iron has an oxidation number of +3, i.e. the iron is present as Fe3+ ions. (For contrast iron(II) oxide would be FeO, with Fe2+). [from: Oxidation number (oxidation state): a number given to an element to show whether it has been oxidised or reduced; for a simple ion it is the charge on the ion (Fe2+ has oxidation number +2).]

From your chapter: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; Redox reaction: a reaction involving simultaneous reduction and oxidation.; OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.; Oxidation number (oxidation state): a number given to an element to show whether it has been oxidised or reduced; for a simple ion it is the charge on the ion (Fe2+ has oxidation number +2).; C6.03 Redox reactions; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.

Question 18. *[Combined Science 2025 21]*

15 A reaction pathway diagram for a reaction is shown.



Which statement describes and explains energy change X?

A Thermal energy is given out as bonds break.

B Thermal energy is given out as bonds form.

C Thermal energy is taken in as bonds break.

D Thermal energy is taken in as bonds form.

Final answer: B (thermal energy is given out as bonds form)

1. On a reaction pathway diagram, energy is plotted up the y-axis and progress of reaction along the x-axis. The reactants start on a flat line on the left, the curve rises to a peak (the activation energy 'hump'), and then falls to another flat line on the right for the products. In this diagram the products line is BELOW the reactants line, so the reaction gives out energy overall - it is exothermic. [from: C6.02 Rate of reaction]

2. The arrow labelled X starts at the top of the hump and points DOWN to the products line. So X is the energy change as the reaction goes from the top of the hump (where old bonds have just been broken) down to the products (where new bonds have just formed). [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

3. Going UP to the top of the hump takes energy in, because the old bonds in the reactants have to be broken first. Breaking bonds always needs energy taken in. So the uphill part is bond breaking and takes energy in - X is NOT that part. [from: C6.01 Physical and chemical changes]

4. Coming DOWN from the top of the hump to the products gives energy out, because the new bonds in the products are forming. Forming bonds always releases energy (gives out thermal energy). X is this downhill arrow, so X represents thermal energy being given out while new bonds form in the products. [from: C6.01 Physical and chemical changes]

5. Match this to the options. A (given out as bonds break) - wrong, bond breaking takes energy in. B (given out as bonds form) - correct. C (taken in as bonds break) - correct idea about bond breaking, but that is the uphill part, not X. D (taken in as bonds form) - wrong, bond forming gives energy out. So the answer is B. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.

Question 19. *[Combined Science 2025 33]*

Use words from the list.

Each word may be used once, more than once or not at all.

anions **cations** **covalent** **electrons** **ionic**

neutrons **neutralisation** **oxidation** **protons** **reduction**

During the reaction, magnesium atoms lose ..... to form positive ions.

These positive ions are known as ..... .

Oxygen atoms from the water molecules form negative ions.

Positive ions and negative ions attract each other and form ..... bonds.

During the reaction, magnesium atoms gain oxygen. This type of reaction is ..... .

[4]

(b) The student repeats the experiment using copper instead of magnesium.

Predict what the student observes during the reaction.

Explain your answer.

observation .....

.....

explanation .....

.....

[2]

[Total: 9]

Final answer: (a)(iii) electrons; cations; ionic; oxidation. (b) observation: no reaction / no visible change / no

1. (a)(iii) First blank. Magnesium atoms lose particles to form positive ions (Mg -> Mg2+ + 2e-). Using OIL RIG, oxidation is loss of electrons, so the particles lost are ELECTRONS. [from: OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.]

2. (a)(iii) Second blank. A positive ion is called a CATION (anions are negative). So Mg2+ is a cation. [from: C6.03 Redox reactions (key terms: oxidation number, ionic, cation vs anion terminology)]

3. (a)(iii) Third blank. When a positive metal ion (Mg2+) and a negative oxide ion (O2-) attract each other, the bond formed is an IONIC bond (electrostatic attraction between oppositely charged ions). [from: C6.03 Redox reactions (ionic equation; positive and negative ions)]

4. (a)(iii) Fourth blank. Gaining oxygen is the oxygen-view definition of oxidation, so the reaction type is OXIDATION. Mg + H2O -> MgO + H2; magnesium gains oxygen to form magnesium oxide. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

5. (b) Observation. Copper is a much less reactive metal than magnesium (copper sits below hydrogen in the reactivity series, magnesium sits well above it). So with steam the student sees NO reaction: no bubbles of gas, no colour change on the copper, and the copper is not used up. [from: Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.]

6. (b) Explanation. Copper is LESS REACTIVE than magnesium, so copper is not reactive enough to pull the oxygen off the water molecules / to displace hydrogen from steam. Therefore no new substances are formed and no hydrogen gas is produced. [from: C6.03 Redox reactions (reactivity / displacement: more reactive metal displaces less reactive)]

From your chapter: OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.; C6.03 Redox reactions

Question 20. *[Combined Science 2025 23]*

A addition of carbon dioxide to limewater

B addition of sodium to water

C electrolysis of molten lead(II) bromide

D separation of petroleum by fractional distillation

Final answer: D (separation of petroleum by fractional distillation — no new substances are formed, it is jus

1. Recall the rule. A physical change only alters the state, form or situation of a substance (e.g. melting, boiling, evaporating, dissolving, separating a mixture) and does NOT produce a new substance. A chemical change produces one or more new substances. [from: C6.01 Physical and chemical changes]

2. Test option A: adding carbon dioxide to limewater. Limewater turns milky because calcium carbonate (a new solid) is formed from the CO2 and the calcium hydroxide in the limewater. A new substance is formed, so this is a CHEMICAL change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

3. Test option B: adding sodium to water. Sodium metal reacts vigorously with water, fizzing and producing hydrogen gas plus sodium hydroxide solution. New substances are formed, so this is a CHEMICAL change. [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

4. Test option C: electrolysis of molten lead(II) bromide. Electricity breaks the compound PbBr2 down into lead metal at one electrode and bromine gas at the other. Two new substances are formed, so this is a CHEMICAL change (a decomposition). [from: Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.]

5. Test option D: separating petroleum by fractional distillation. Petroleum is a mixture; fractional distillation just boils off and condenses the different hydrocarbons at their different boiling points. No new substance is made — the same molecules are just separated. Therefore this is a PHYSICAL change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

6. Only option D does not form a new substance, so the answer is D. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.

Question 21. *[Combined Science 2025 23]*

Which change decreases the rate of this reaction?

A adding a catalyst to the reaction mixture

B increasing the concentration of the hydrochloric acid

C increasing the size of the zinc lumps

D increasing the temperature of the hydrochloric acid

18 The equation for the complete combustion of ethane is shown.

Final answer: C (increasing the size of the zinc lumps decreases the surface area in contact with the acid,

1. List the five factors that change the rate of a reaction: surface area of solid, concentration of solution, pressure of gases, temperature, and presence of a catalyst. The question asks which change DECREASES the rate. [from: C6.02 Rate of reaction]

2. Check option A: adding a catalyst. A catalyst speeds a reaction up, so rate INCREASES. Not the answer. [from: Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.]

3. Check option B: increasing the concentration of HCl. More acid particles per unit volume means more frequent collisions with the zinc surface, so rate INCREASES. Not the answer. [from: C6.02 Rate of reaction]

4. Check option C: increasing the SIZE of the zinc lumps. Bigger lumps have a SMALLER total surface area exposed to the acid for the same mass, so there are fewer collision sites per second between H+ ions and Zn. Rate DECREASES. This is the answer. [from: C6.02 Rate of reaction]

5. Check option D: increasing the temperature. Higher temperature means particles have more kinetic energy, collide more often, and a greater proportion of collisions exceed Ea, so rate INCREASES. Not the answer. [from: Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).]

6. Only option C decreases the rate, so the answer is C. [from: C6.02 Rate of reaction]

From your chapter: C6.02 Rate of reaction; Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.; Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.; Higher temperature increases reaction rate because (a) particles move faster so collide more often and (b) a greater proportion of collisions exceed Ea (this second effect dominates).

Question 22. *[Combined Science 2025 23]*

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$$

Which statement about this reaction is correct?

A Ethane is oxidised because it gains oxygen.

B Ethane is reduced because carbon–hydrogen bonds are broken.

C It is endothermic because thermal energy is transferred to the surroundings.

D It is exothermic because thermal energy is taken in from the surroundings.

Final answer: A Ethane is oxidised because it gains oxygen

1. Look at what happens to ethane (C2H6) in the equation 2C2H6 + 7O2 -> 4CO2 + 6H2O. The carbon atoms in ethane go from being bonded only to hydrogen to being bonded to oxygen (in CO2). So ethane has gained oxygen during the reaction. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

2. The chapter profile says 'oxidation is the gain of oxygen'. Because ethane gains oxygen, ethane is oxidised. That makes statement A correct. [from: C6.03 Redox reactions]

3. Check statement B. Reduction is the loss of oxygen (or the gain of electrons), NOT simply 'breaking C-H bonds'. Bonds are broken in every reaction; that is not what defines reduction. So B is wrong. [from: Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.]

4. Check statements C and D, which are about energy. Combustion (burning) is always exothermic - it gives out thermal energy to the surroundings, which is why fuels are useful. 'Exothermic' means energy is given out, 'endothermic' means energy is taken in. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

5. Statement C says the reaction is endothermic because energy is transferred TO the surroundings - that is a contradiction (transferring energy to the surroundings is exothermic, not endothermic). So C is wrong. Statement D says it is exothermic because energy is taken IN from the surroundings - that is also a contradiction (taking energy in is endothermic). So D is wrong too. The only correct statement is A. [from: C6.01 Physical and chemical changes]

From your chapter: C6.01 Physical and chemical changes; C6.03 Redox reactions; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.

Question 23. *[Combined Science 2025 12]*

17 Which process is a physical change?

A electrolysis

B melting

C oxidation

D polymerisation

Final answer: B (melting)

1. A physical change only alters the physical state or form of a substance. No new substance is made and it is usually easy to reverse. A chemical change (chemical reaction) makes at least one new substance and is usually hard to reverse. So I need the option where nothing new is made. [from: C6.01 Physical and chemical changes]

2. Option A electrolysis: an electric current splits a compound into new substances (for example copper and chlorine from copper(II) chloride). New substances are formed, so this is a chemical change. [from: C6.01 Physical and chemical changes]

3. Option B melting: a solid turns into a liquid (for example ice to water). Only the physical state changes, the substance is still the same, and it can be reversed by freezing. This is a physical change. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

4. Option C oxidation: a substance gains oxygen (or loses electrons), for example magnesium burning in oxygen to form magnesium oxide. A new substance is made, so this is a chemical change. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

5. Option D polymerisation: small molecules (monomers) join together to form a new, long-chain molecule (polymer). A new substance is made, so this is a chemical change. [from: C6.01 Physical and chemical changes]

6. Only melting fits the definition of a physical change, so the answer is B. [from: Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.]

From your chapter: C6.01 Physical and chemical changes; Physical change: a change in the physical state, form or appearance of a substance that does not produce any new chemical substance.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.

Question 24. *[Combined Science 2025 41]*

(c) Fig. 4.2 is a reaction pathway diagram for a reaction catalysed by copper.

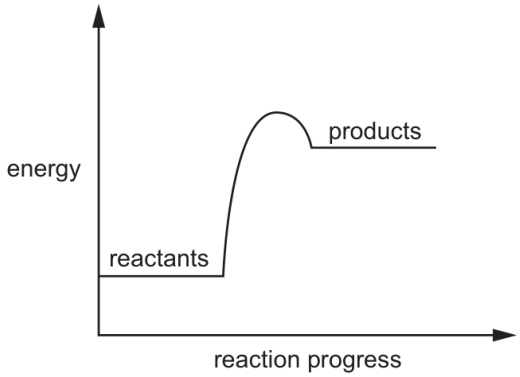


Fig. 4.2

(i) Identify the type of chemical reaction shown in Fig. 4.2.

Explain your answer.

type of chemical reaction .....

explanation .....

.....

..... [2]

(ii) Draw an arrow on Fig. 4.2 to show the activation energy for this reaction. [1]

(iii) Define activation energy.

.....

.....

.....

..... [2]

(iv) Describe what is meant by a catalyst.

.....

.....

.....

..... [2]

[Total: 11]

Final answer: (i) Endothermic (products higher than reactants on the pathway diagram). (ii) Arrow from the

1. (c)(i) Read the pathway diagram: reactants start at a low energy, the curve rises over an activation-energy hump, and the products settle at an energy LEVEL ABOVE the reactants. Because products have more stored energy than reactants, energy must have been taken in from the surroundings. This matches the definition of an endothermic reaction. [from: C6.01 Physical and chemical changes (exothermic/endothermic)]

2. (c)(i) State the answer and explanation. Type of reaction: endothermic. Explanation: on the reaction pathway diagram the products are at a higher energy level than the reactants, so energy has been absorbed (taken in) from the surroundings. [from: Physical change / chemical change definitions]

3. (c)(ii) Activation energy on a reaction pathway diagram is the height of the hump measured from the reactants' energy line up to the peak. Draw a vertical double-headed arrow that starts on the horizontal 'reactants' line and ends exactly at the top of the curve (the peak/transition state). Label it Ea. [from: C6.02 Rate of reaction (reaction pathway diagram, activation energy)]

4. (c)(iii) Define activation energy using the chapter profile definition. Activation energy (Ea) is the MINIMUM energy that colliding reactant particles must have in order for a reaction to occur. Particles with less than this energy simply bounce off each other without reacting. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

5. (c)(iv) Define catalyst using the chapter profile definition. A catalyst is a substance that INCREASES the rate of a chemical reaction by providing an alternative pathway with a LOWER activation energy, and is itself CHEMICALLY UNCHANGED at the end of the reaction. Two mark-bearing ideas: (a) speeds up the reaction / lowers Ea; (b) is not used up / is chemically unchanged at the end. [from: Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Catalyst: a substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation energy and is itself chemically unchanged at the end of the reaction.; reaction pathway diagram

Question 25. *[Combined Science 2025 62]*

2 A student investigates the reaction between aluminium and aqueous copper(II) chloride.

Aqueous copper(II) chloride is a blue solution.

Aluminium is a grey solid.

(a) Procedure

The student:

- assembles the apparatus shown in Fig. 2.1

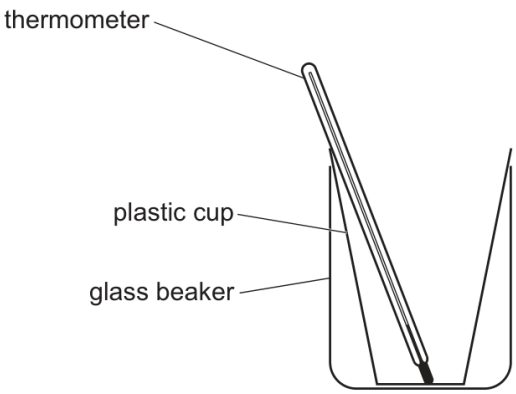


Fig. 2.1

- pours 25 cm^3 of aqueous copper(II) chloride into the plastic cup
- records in Table 2.1, the initial temperature of the aqueous copper(II) chloride
- adds 0.5 g of aluminium to the aqueous copper(II) chloride
- stirs the mixture in the plastic cup
- records in Table 2.1, the highest temperature reached by the reaction mixture.

(i) Fig. 2.2 shows the temperature readings for this experiment.

Fig. 2.2

Record in Table 2.1, these temperatures to the nearest 0.5 °C.

Final answer: (a)(i) initial temperature = 22.0 degrees C; highest temperature reached = 41.0 degrees C

1. (a)(i) Read the left-hand thermometer in Fig 2.2 (labelled 'initial temperature'). The scale shows 10, 20 and 30 degrees C with small divisions between each 10 degree mark. The top of the coloured liquid sits just above the 20 mark. Read to the nearest 0.5 degrees C: initial

temperature = 22.0 degrees C. [from: C6.01 Physical and chemical changes]

2. (a)(i) Read the right-hand thermometer (labelled 'highest temperature reached'). The scale shows 30, 40 and 50 degrees C. The top of the coloured liquid sits just above the 40 mark. Read to the nearest 0.5 degrees C: highest temperature reached = 41.0 degrees C. [from: C6.01 Physical and chemical changes]

3. (a)(i) Sanity check: the temperature has gone up from 22.0 to 41.0 degrees C, a rise of 19 degrees C. This makes sense because aluminium reacting with aqueous copper(II) chloride is an exothermic displacement reaction - aluminium is more reactive than copper and displaces copper metal, giving out thermal energy to the solution. A clear temperature rise is exactly what we expect. [from: Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.]

From your chapter: C6.01 Physical and chemical changes; Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.; Chemical change (chemical reaction): a change in which one or more new substances are formed; usually difficult to reverse and accompanied by an energy change.

Question 26. *[Combined Science 2025 31]*

12

6 **(a)** Fig. 6.1 shows the combustion of magnesium ribbon in air.

A white powder of magnesium oxide is formed.

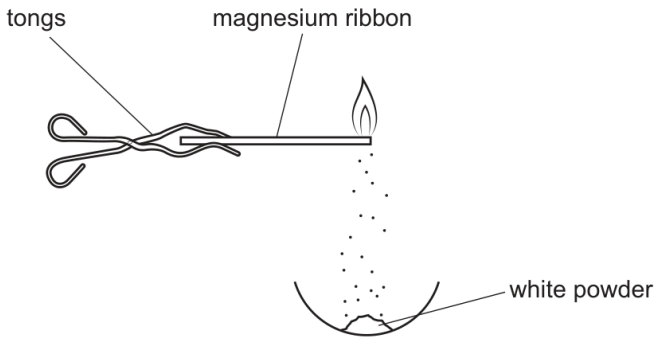


Fig. 6.1

(i) Name the gas that reacts with the magnesium.

..... [1]

(ii) Magnesium oxide is a metallic oxide.

Circle the term that describes this type of oxide.

acidic **aqueous** **basic** **carbonate** **solvent**

[1]

(iii) This reaction is exothermic.

Describe what is meant by exothermic.

.....

..... [1]

Final answer: (a)(i) oxygen; (a)(ii) basic; (a)(iii) exothermic means heat (energy) is given out to the surroun

1. (a)(i) Magnesium is burning in air. Air is a mixture of gases, but the gas in air that reacts chemically with magnesium is oxygen. The equation is 2Mg(s) + O2(g) -> 2MgO(s), so the gas that reacts with magnesium is oxygen (O2). This fits the chapter example of combustion and the definition of combustion (reaction with oxygen). [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

2. (a)(ii) Magnesium is a metal, so magnesium oxide is a metal oxide. Metal oxides are basic oxides (they react with acids to form a salt and water). So from the list (acidic, aqueous, basic, carbonate, solvent) the correct term is basic. [from: C6.01 Physical and chemical changes]

3. (a)(iii) Exothermic means the reaction gives out energy (heat) to the surroundings, so the surroundings get hotter. In the combustion of magnesium you see a bright white flame and feel heat, which shows that energy is released. An answer like 'heat (energy) is given out/released to the surroundings' scores the mark. [from: During a chemical change energy can be given out (exothermic) or taken in (endothermic).]

From your chapter: C6.01 Physical and chemical changes; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.; During a chemical change energy can be given out (exothermic) or taken in (endothermic).

Question 27. *[Combined Science 2025 21]*

19 Which metal reacts most quickly with cold water?

A calcium

B magnesium

C potassium

D sodium

Final answer: C — potassium

1. List what each option is. A = calcium (Group 2), B = magnesium (Group 2), C = potassium (Group 1 alkali metal), D = sodium (Group 1 alkali metal). The question asks which reacts MOST QUICKLY (i.e. highest rate of reaction) with COLD water. [from: C6.02 Rate of reaction (rate of reaction = change in amount of reactant or product / change in time)]

2. Use the reactivity idea from the chapter: a more reactive metal reacts faster and more vigorously than a less reactive metal (this is the same idea used in displacement reactions, where the more reactive metal pushes the less reactive one out of its solution). Order of reactivity with cold water (most to least): potassium > sodium > calcium > magnesium. [from: Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.]

3. Apply the reactivity order to cold water. Potassium and sodium are Group 1 alkali metals and react vigorously with cold water, producing hydrogen gas and an alkaline hydroxide solution. Calcium (Group 2) also reacts with cold water but more slowly. Magnesium reacts only VERY slowly with cold water (it needs steam to react quickly). So the fastest of the four with cold water is potassium. [from: C6.02 Rate of reaction (speed of reaction)]

4. Compare the two most reactive options, potassium and sodium. Potassium is below sodium in Group 1 of the Periodic Table, so potassium is MORE reactive than sodium. Potassium reacts so quickly with cold water that the heat released (exothermic) ignites the

hydrogen produced (you see a lilac flame). Sodium also reacts quickly but doesn't usually ignite. So potassium reacts most quickly of the four. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

5. Select the letter. Potassium is option C. Final answer: C — potassium. [from: C6.02 Rate of reaction]

From your chapter: C6.02 Rate of reaction; rate of reaction = change in amount of reactant or product / change in time; Displacement reaction: a reaction in which a more reactive element displaces a less reactive element from a compound or solution.; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.

Question 28. *[Combined Science 2025 43]*

4 A student investigates the rate of reaction between large pieces of magnesium carbonate, MgCC and dilute hydrochloric acid, HC*l*, using the apparatus shown in Fig. 4.1.

The student records the time taken to produce 10 cm^3 of gas.

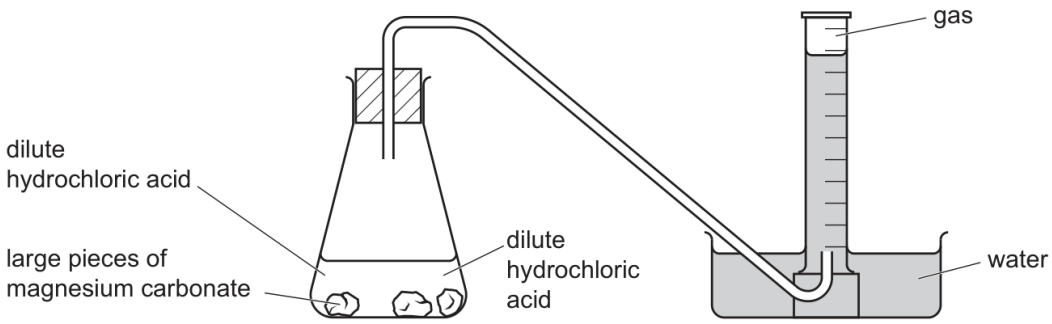


Fig. 4.1

(a) Three products are formed in the reaction: magnesium chloride $MgCl_2$, a gas and one oth product.

Give the balanced symbol equation for this reaction.

..... [

(b) The student repeats the experiment using:

- small pieces of magnesium carbonate
- magnesium carbonate powder.

All other conditions are kept constant.

Table 4.1 shows the time taken to produce 10 cm^3 of gas and the rate of reaction.

Table 4.1

magnesium carbonate	time taken to produce 10 cm^3 of gas in s	rate of reaction in cm^3 per s
large pieces	15.2	0.66
small pieces	8.3	1.2
powder	5.1	

(i) Calculate the rate of reaction in cm^3 per s for magnesium carbonate powder.

Write your answer in Table 4.1. [

Final answer: Smaller pieces of magnesium carbonate make the reaction faster, because for the same ma

1. (b.ii) State the effect: if the magnesium carbonate is used as smaller pieces (or a powder) instead of large lumps, the rate of reaction is faster. Bigger pieces react more slowly. [from: C6.02 Rate of reaction]

2. (b.ii) Explain using surface area: for the same mass of magnesium carbonate, smaller pieces have a larger total surface area exposed to the acid (think of breaking one big cube into many small cubes — the outside area goes up). [from: C6.02 Rate of reaction (surface area factor)]

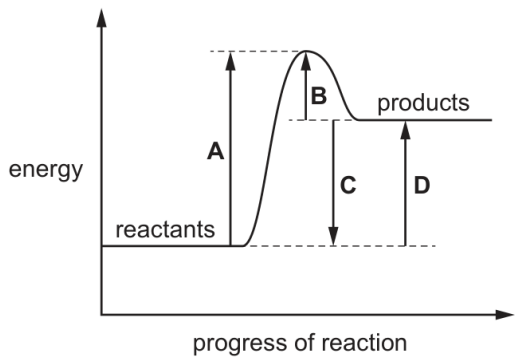
3. (b.ii) Explain using collision theory: a reaction only happens when acid particles collide with magnesium carbonate particles on the surface with enough energy. A larger surface area means acid particles can hit the solid in more places, so there are more frequent successful collisions per second, so the reaction is faster. [from: Collision theory: a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.]

4. (c) Note on scope: this sub-part asks for a dot-and-cross diagram of the ionic bonding in magnesium chloride. That is a bonding topic, not a Chemical Reactions topic, so it is covered in chapter C9 (Bonding), not in C6. It is therefore not solved here. [from: out_of_scope_notes]

From your chapter: C6.02 Rate of reaction; surface area; collision theory; frequency of collisions; successful collision; Collision theory: a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.

Question 29. *[Combined Science 2025 23]*

19 Which arrow on the reaction pathway diagram represents the overall energy change for the reaction?



Final answer: D (the arrow from the reactants-level dashed line up to the products line shows the overall en

1. Look at the reaction pathway diagram. The left flat line is labelled 'reactants' and sits at a lower energy. The curve goes up and over a hump (the peak) and comes down to a flat line on the right labelled 'products' which sits at a higher energy than reactants. Because the products are higher than the reactants, the overall reaction is endothermic (energy is taken in from the surroundings). [from: C6.01 Physical and chemical changes (endothermic: products higher than reactants)]

2. Recall what 'overall energy change' means on a reaction pathway diagram. The overall energy change (also called delta H) is the vertical distance between the reactants line and the products line. For an endothermic reaction it is measured from the reactants level UP to the products level. It is NOT the hump, and it is NOT measured from the x-axis. [from: C6.01 Physical and chemical changes (exothermic vs endothermic energy changes)]

3. Check each labelled arrow on the diagram: A goes from the x-axis up to the reactants line - this is just the energy of the reactants above a zero reference, NOT the overall energy change. B goes from the reactants line up to the top of the peak - this is the activation energy (Ea), NOT the overall energy change. C goes from the x-axis all the way up to the top of the peak - this is the total energy of the peak from zero, NOT the overall energy change. D goes

from the reactants-level dashed line up to the products line - this IS the overall energy change (delta H) for the endothermic reaction. [from: C6.02 Rate of reaction (reaction pathway diagram labels: Ea vs overall energy change)]

4. Only arrow D measures the vertical distance between the reactants line and the products line, which is the overall energy change for the reaction. So the answer is D. [from: C6.01 Physical and chemical changes (endothermic energy change = products - reactants)]

From your chapter: C6.01 Physical and chemical changes; C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; reaction pathway diagram; activation energy; Ea; endothermic; exothermic

Question 30. *[Combined Science 2025 42]*

(c) The reaction between aluminium and iron(III) oxide is exothermic.

A flame is used to provide the activation energy to start the reaction.

Fig. 5.1 shows the incomplete reaction pathway diagram for this reaction.

Complete Fig. 5.1.

Include:

- the energy level of the products
- a labelled arrow to show the activation energy, E_a
- a labelled arrow to show the overall energy change of the reaction.

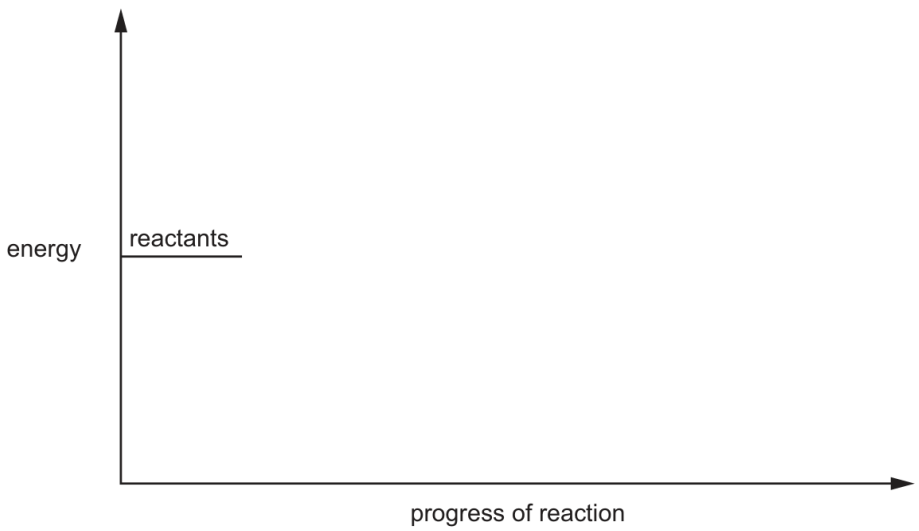


Fig. 5.1

[3]

[Total: 9]

Final answer: On the diagram: draw a horizontal PRODUCTS line BELOW the reactants line (reaction is ex

1. (c) The reaction is EXOTHERMIC: it releases energy to the surroundings. On a reaction pathway diagram that means the products must sit LOWER on the energy axis than the reactants. So first, draw a horizontal line labelled 'products' below the existing reactants line. [from: C6.02 Rate of reaction (reaction pathway diagram; exothermic reactions release energy so products are lower than reactants)]

2. (c) Draw the reaction pathway itself: a smooth curve that starts at the left-hand end of the reactants line, rises up to a peak (the 'hump'), then falls down to the right-hand end of the products line. This hump is the energy barrier the particles must climb before they can react. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

3. (c) Label Ea (activation energy). Draw a VERTICAL arrow on the left of the hump that goes from the REACTANTS line UP to the TOP of the hump. Label it Ea. This is the extra energy the flame has to supply to get the reaction started. [from: Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.]

4. (c) Label the overall energy change. Draw a second VERTICAL arrow on the right-hand side of the diagram that goes from the REACTANTS line DOWN to the PRODUCTS line. Label it 'overall energy change' (or Delta H, energy released). Because the products are lower, this arrow points downward, which tells you energy has been released - exactly what 'exothermic' means. [from: Collision theory / C6.02 reaction pathway diagram; exothermic energy change]

5. (c) Sanity check. The flame only has to supply enough energy to lift reactants to the top of the hump (Ea). Once the reaction starts, it releases more energy than Ea (the overall energy change), so the reaction then keeps itself going - that is why the thermite reaction carries on fiercely once lit. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

From your chapter: C6.02 Rate of reaction; Activation energy (Ea): the minimum energy that colliding particles must have for a reaction to occur.; Collision theory: the idea that a chemical reaction takes place only when reactant particles collide with each other with an energy at least equal to the activation energy.; reaction pathway diagram

Question 31. *[Combined Science 2025 12]*

21 Which two substances form when magnesium reacts with steam?

A magnesium hydroxide and hydrogen

B magnesium hydroxide and oxygen

C magnesium oxide and hydrogen

D magnesium oxide and water

22 Anhydrous copper(II) sulfate is used to test for water.

Final answer: C — magnesium oxide and hydrogen

1. Write the word equation for magnesium reacting with steam. Steam is hot water vapour (H2O). The equation is: magnesium + steam -> magnesium oxide + hydrogen. Symbol equation: Mg(s) + H2O(g) -> MgO(s) + H2(g). So the two products are magnesium oxide and hydrogen gas. [from: C6.03 Redox reactions]

2. Use the oxygen view of redox to check the answer. Magnesium gains an oxygen atom (Mg -> MgO) so magnesium is oxidised. Water loses its oxygen (H2O -> H2) so water is reduced — hydrogen gas is released. This confirms the products are magnesium oxide and hydrogen. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

3. Compare to the options. A says magnesium hydroxide + hydrogen — magnesium only forms hydroxide with cold liquid water, not with steam. B says magnesium hydroxide + oxygen — wrong, no oxygen is released. D says magnesium oxide + water — wrong, steam is the reactant not a product. C (magnesium oxide + hydrogen) matches our equation, so C is correct. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

From your chapter: C6.03 Redox reactions; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.

Question 32. *[Combined Science 2025 23]*

23 The equations describe the reactions that happen when iron is extracted in the blast furnace.

$$C + O_2 \rightarrow CO_2$$

$$C + CO_2 \rightarrow 2CO$$

$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

What is a description of one of these reactions?

A Carbon dioxide is oxidised to form carbon monoxide.

B Iron(III) oxide is reduced by carbon dioxide.

C Carbon reacts with carbon monoxide to form carbon dioxide.

D Carbon monoxide removes oxygen from iron(III) oxide.

24 The structures of some compounds are shown.

Final answer: D. Carbon monoxide removes oxygen from iron(III) oxide.

1. Use the oxygen view of redox: oxidation = gain of oxygen, reduction = loss of oxygen. Check each answer against the three blast-furnace equations given. [from: C6.03 Redox reactions]

2. Option A: 'Carbon dioxide is oxidised to form carbon monoxide.' Equation 2 is C + CO2 -> 2CO. CO2 has LOST oxygen (CO2 -> CO), so CO2 is REDUCED, not oxidised. A is wrong. [from: Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.]

3. Option B: 'Iron(III) oxide is reduced by carbon dioxide.' Equation 3 is Fe2O3 + 3CO -> 2Fe + 3CO2. Fe2O3 is reduced by CO (carbon monoxide), NOT by carbon dioxide. B is wrong. [from: Reduction of zinc oxide by carbon: ZnO(s) + C(s) -> Zn(s) + CO(g). Zinc is reduced; carbon is oxidised.]

4. Option C: 'Carbon reacts with carbon monoxide to form carbon dioxide.' The real equation 2 is C + CO2 -> 2CO (carbon reacts with carbon DIOXIDE to form carbon MONOXIDE). C describes the reverse direction, so C is wrong. [from: C6.03 Redox reactions]

5. Option D: 'Carbon monoxide removes oxygen from iron(III) oxide.' Equation 3 is Fe2O3 + 3CO -> 2Fe + 3CO2. Fe2O3 loses oxygen (Fe2O3 -> Fe), CO gains oxygen (CO -> CO2). So CO does remove oxygen from Fe2O3 (Fe2O3 is reduced; CO is oxidised). D is the correct description. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

From your chapter: C6.03 Redox reactions; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; OIL RIG mnemonic: Oxidation Is Loss of electrons, Reduction Is Gain of electrons.

Question 33. *[Combined Science 2025 32]*

4 A student investigates the reaction between magnesium and steam.

Fig. 4.1 is a diagram of the apparatus used in the investigation.

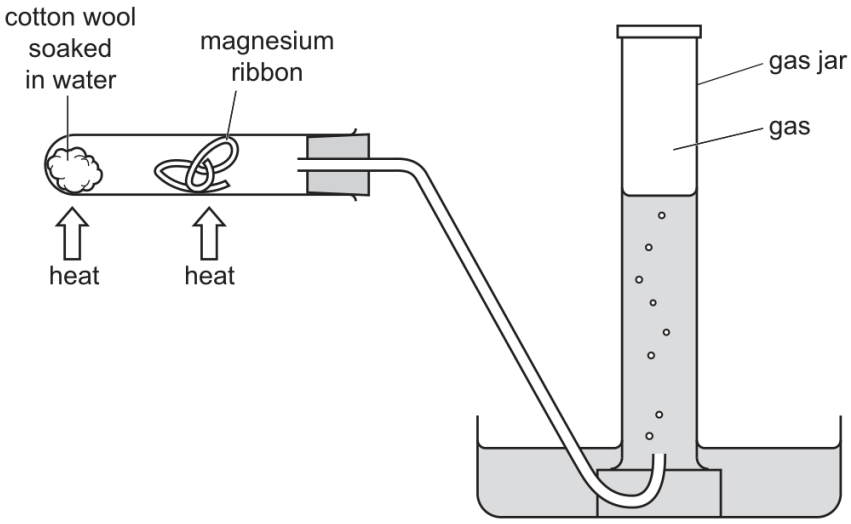


Fig. 4.1

(a) The reaction produces magnesium oxide and a gas.

(i) Complete the word equation for this reaction.

	+		→	magnesium oxide	+	

[

(ii) The gas jar in Fig. 4.1 collects the gas, but it does **not** allow the volume to b measured directly.

Name **one** piece of apparatus that allows the volume of gas to be measured directly.

..... [

Final answer: (a)(i) magnesium + steam (water) -> magnesium oxide + hydrogen; (a)(ii) gas syringe; (a)(iii)

1. (a)(i) The cotton wool soaked in water gives off steam when the tube is heated. The steam passes over the hot magnesium ribbon and reacts with it. Magnesium gains oxygen from the water to form magnesium oxide; the hydrogen from the water is set free as hydrogen gas. Word equation: magnesium + steam -> magnesium oxide + hydrogen. (Writing 'water' instead of 'steam' is also accepted.) Symbol equation: Mg(s) + H2O(g) -> MgO(s) + H2(g). 1 mark for 'magnesium' and 'steam/water'; 1 mark for 'hydrogen'. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

2. (a)(ii) A gas jar just traps the gas but has no volume scale, so it cannot measure volume directly. The piece of apparatus listed in the chapter that measures gas volume directly is a gas syringe. (An inverted measuring cylinder filled with water and held over the delivery tube is also accepted, but the gas syringe is the clearest answer.) [from: Rates are investigated by measuring loss in mass on a balance, volume of gas collected in a gas syringe, or time for a cross to disappear]

3. (a)(iii) First blank: 'magnesium atoms lose ____ to form positive ions'. OIL RIG tells us oxidation is the loss of electrons, so a Mg atom loses electrons to form Mg^{2+}. Answer: electrons. [from: Oxidation (electron view): the loss of electrons or an increase in oxidation number during a reaction.]

4. (a)(iii) Second blank: 'these positive ions are known as ____'. Positive ions (Fe^{2+}, Cu^{2+}, Zn^{2+}, Mg^{2+}) are called cations. Answer: cations. [from: Roman numerals show oxidation numbers of transition metals, e.g. iron(II) is Fe^{2+}]

5. (a)(iii) Third blank: 'positive and negative ions attract each other and form ____ bonds'. The type of bond between oppositely charged ions is an ionic bond. Answer: ionic. [from: ionic equation]

6. (a)(iii) Fourth blank: 'magnesium atoms gain oxygen; this type of reaction is ____'. Gain of oxygen is the oxygen-view definition of oxidation. Answer: oxidation. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

7. (b) If copper is used instead of magnesium, the student will see no (visible) reaction — no bubbles of gas, no glow, no change in the copper, and no gas collected in the gas jar. [from: C6.03 Redox reactions]

8. (b) The explanation uses the idea of reactivity from the chapter. In the chapter we saw that copper(II) oxide can be reduced by hydrogen: $CuO + H_2 \rightarrow Cu + H_2O$. That means hydrogen is more reactive than copper, so copper cannot take oxygen away from water/steam. Copper is also less reactive than magnesium, so it does not react with steam — no magnesium oxide equivalent is formed and no hydrogen gas is produced. 1 mark for the observation, 1 mark for the reactivity/redox explanation. [from: Reduction of copper(II) oxide by hydrogen: $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$. Copper is reduced; hydrogen is oxidised.]

From your chapter: C6.03 Redox reactions; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; Oxidation (electron view): the loss of electrons or an increase in oxidation number during a reaction.; Roman numerals show oxidation numbers of transition metals, e.g. iron(II) is Fe^{2+}; ionic equation; Reduction of copper(II) oxide by hydrogen: $CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l)$. Copper is reduced; hydrogen is oxidised.; Rates are investigated by measuring loss in mass on a balance, volume of gas collected in a gas syringe, or time for a cross to disappear

Question 34. *[Combined Science 2025 33]*

6 Scientists are investigating the use of iron as a fuel in cars.

Iron reacts with oxygen, as shown in equation 1. This reaction releases energy.

The iron oxide formed is converted back to iron, as shown in equation 2.

equation 1 .....$Fe(s)$ + ...O_2(.....) $\rightarrow$ $2Fe_2O_3(s)$

equation 2 $Fe_2O_3(s) + 3H_2(g) \rightarrow 2Fe(s) + 3H_2O(l)$

(a) Balance equation 1 and add the missing state symbol for oxygen. [

(b) State the name given to a chemical reaction that releases thermal energy.

..... [

(c) Use the substances in equation 1 and equation 2 to answer the following questions.

(i) Identify the compound that exists as simple molecules.

..... [

(ii) Identify the transition element.

..... [

(d) Explain why equation 1 shows that iron is oxidised.

.....

..... [

(e) Most cars burn fuels that contain carbon.

Explain why using iron as a fuel may cause less harm to the environment than using fue that contain carbon.

.....

.....

.....

..... [

[Total:

Final answer: (a) 4Fe(s) + 3O2(g) -> 2Fe2O3(s); (b) exothermic; (c)(i) H2O (water); (c)(ii) Fe (iron); (d) iron

1. (a) Balance __Fe(s) + __O2(?) -> 2Fe2O3(s). The RHS has 2 x Fe2O3 = 4 Fe atoms and 6 O atoms. To match, the LHS needs 4 Fe atoms and 6 O atoms. 6 O atoms come from 3 O2 molecules (3 x 2 = 6). So the balanced equation is 4Fe(s) + 3O2(g) -> 2Fe2O3(s). [from: C6.01 Physical and chemical changes]

2. (a) The missing state symbol for O2 is (g) - oxygen is a gas at room temperature. [from: C6.01 Physical and chemical changes]

3. (b) A chemical reaction that releases thermal (heat) energy to the surroundings is called EXOTHERMIC. Burning iron gives out heat energy, so this is an exothermic reaction. [from: Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.]

4. (c)(i) A compound is a substance made of two or more different elements chemically joined. In the two equations the compounds present are Fe_2O_3 and H_2O (Fe, O_2 and H_2 are elements, not compounds). Fe_2O_3 is a giant ionic solid. H_2O (water) is made of small H_2O molecules held together by weak forces, so water exists as simple molecules. Answer: H_2O (water). [from: C6.01 Physical and chemical changes]

5. (c)(ii) Transition elements are the block of metals in the middle of the Periodic Table (and they often use Roman-numeral oxidation numbers). The only element in the equations from that block is iron (Fe). Answer: Fe (iron). [from: C6.03 Redox reactions]

6. (d) Use the oxygen view of redox. In equation 1, iron (Fe) starts on its own and ends up combined with oxygen as Fe_2O_3. Iron has GAINED oxygen. Gain of oxygen is the definition of oxidation, so iron is oxidised. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

7. (e) Point 1 (harm from carbon fuels): When fuels that contain carbon burn, they produce carbon dioxide (CO_2). CO_2 is a greenhouse gas, so it adds to global warming / climate change. (Incomplete combustion can also make toxic carbon monoxide, CO.) [from: Combustion of methane: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (exothermic, redox).]

8. (e) Point 2 (iron fuel is cleaner): When iron burns (equation 1, $4Fe + 3O_2 \rightarrow 2Fe_2O_3$) the only product is iron(III) oxide, which is a solid. No CO_2, no CO and no sulfur dioxide are produced, so burning iron does not add greenhouse gases or acid rain gases to the atmosphere. [from: Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.]

9. (e) Point 3 (iron can be recycled): Equation 2 ($Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$) shows the iron(III) oxide waste can be reduced back to iron. So the same iron can be used as fuel again - iron is a recyclable/sustainable fuel, unlike carbon fuels which are burned once and gone. [from: Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.]

From your chapter: C6.01 Physical and chemical changes; C6.03 Redox reactions; Combustion: a chemical reaction in which a substance reacts with oxygen and releases energy; combustion is exothermic.; Oxidation (oxygen view): the gain of oxygen by an element or compound during a chemical reaction.; Reduction (oxygen view): the loss of oxygen by an element or compound during a chemical reaction.; Combustion of methane: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ (exothermic, redox).

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